

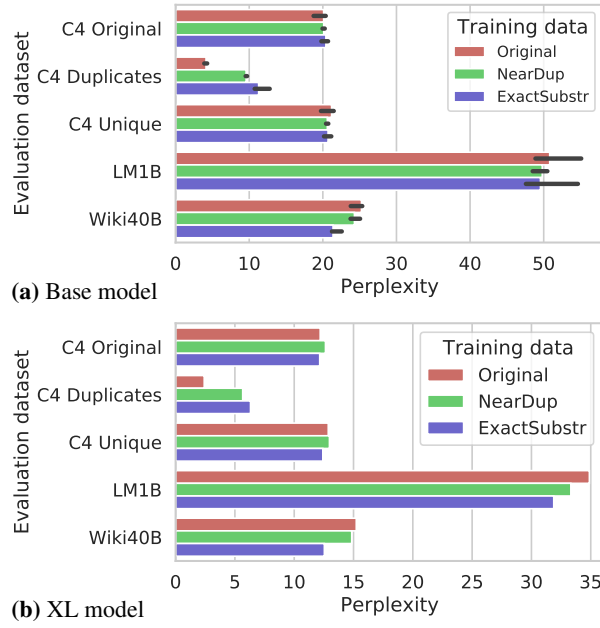


Figure 7: Impact of deduplicating the training set on validation perplexity. In **(a)**, we plot the results from T5 base (110M parameters) across three training runs with different random initializations. The black bar represent the lowest perplexity to the highest perplexity, and the colored bar the median perplexity. In **(b)**, we plot the results from T5 XL (1.5B parameters).

Generated Text	Freq in C4
, you'll need to be knowledgeable to make the very best decisions. We will make sure you know what can be expected. We take the surprises from the picture by giving accurate and thorough information. You can start by talking about your task with our client service staff when you dial 888-353-1299. We'll address all of your questions and arrange the initial meeting. We work closely with you through the whole project, and our team can show up promptly and prepared.	5,497
then Waterside Lodge are well equipped for the task. Our fully equipped family sized lodges offer a comfortable luxurious stay for a fantastic price, giving you beautiful views of the lakes and the surrounding countryside. Offering luxurious self-catering holidays in our fully featured Scandinavian holiday lodges. Perfectly located to explore the beaches, coastline. All of our lodges are sized for 6 people and are furnished to the highest standards to ensure you have a stay like no other. At Waterside Lodge the stay itself is only half of the package, Waterside lodge is situated closely to the Heritage Coast which makes our lodges the perfect stay for anyone wanting to get away and have a relaxing countryside break from the city. Whilst you stay with us be sure to take advantage of all the activities Waterside Lodge has to offer. Such as the use of our on-site fishing lakes for the keen fisherman, free internet access, outside relaxation areas, comfortable lounges and much more.	571
you are only looking to find rent to own homes in your city or are open to exploring all kinds of rent to own home listings, our database does it all. One of the best aspects of iRentToOwn.com is that, besides options to rent to buy a house, it has numerous other categories of home sale options. These include bank foreclosure homes, pre-foreclosure homes, short sales, HUD/government foreclosures, auction homes and owner-financing/FSBO (For Sale By Owner) homes. With help from the convenient search features offered by our site, shoppers are able to find their ideal lease to own home, real estate company, and more in South	51
, IL employs journeyman as licensed to work by themselves, without direct supervision, installing wiring, outlets and fixtures. Our journeyman also does service work, troubleshooting when a breaker fails or a light stops working. Our journeyman does not offer permits that must be issued by our master. Our journeyman follows our master's plans and directions. Our journeyman's responsibilities will vary based on the work that needs to be done. Our journeymen are skilled with residential, commercial and industrial installations and repairs.ust work from six years as an apprentice, under direct supervision of our master, and pass a journeyman test. This person also must have some classroom education on the National Electrical Code and fundamental electricity in a technical school a program affiliated with the National Joint Apprenticeship Training Council. Journeyman training combines hands-on work with education on basic electricity.	6
combustion process of a petrol engine is never perfect. Dangerous gases, such as nitrogen oxide, carbon monoxide and hydrocarbons will arise and it is the job of the catalytic converter to reduce these to safer emissions. These cat converters can fail by becoming clogged, or if the engine has bad exhaust valves or the plugs fail, causing unburned fuel to overheat the converter. Mettam's Mufflers can resolve these issues with your Karr	5
,ANDREW Find the ancestral town: Many a researcher is stuck behind records that say, BIRTHPLACE: IRELAND without saying where in Ireland, or whatever other country. Remember that your immigrant ancestor's siblings probably were born in the same ancestral town, so check all of their records, too. Around 1900, the Roman Catholic churches reported marriages to the churches where the persons were baptised, and before the wedding, they would require a baptismal certificate from that church, without marriage notations, to make sure that the persons were not already married, ordained, or whatever, and were free to marry. Do check the Catholic records especially for ex loco and the home town. If your ancestor's sister had a daughter who generated a marriage or death record saying, MOTHER'S BIRTHPLACE: and the exact town, then you know where to start searching for records that will confirm it is your ancestor's home town. BEWARE: Just because you find a family with the same names does not mean they are the same family, as they could very well be an unrelated family from a different town in the same ancestral country. The webmaster has learned this. One clue was that one family was still having babies in Potenza city, Italy while the other was having babies in Colorado, U.S.A.	2
will not want to search for Power Washing companies in Wyoming on an extensive basis. The service personnel will be at your doorsteps through online or phone booking. The power wash solutions offered by us are matchless and you can compare with others in Winfield, IL. The power wash services offered by us are very economical. Gutter brightener will be applied which will be followed by cleaning through double scrub. The cleaning will be done by using a soft bristle brush. The bond and contaminants will be released in an effortless manner.	1
Z3 Plus are valid in all major cities of India like Delhi, Gurgaon, Noida, Mumbai, Chennai, Bangalore, Hyderabad, Kolkata, Pune, Ahmedabad, Coimbatore, Lucknow, Trichy, Madurai, Trivandrum, Mysore, Jaipur, Chandigarh, Pondicherry, Bhopal, Patna, Bhubaneswar, Amritsar, Cochin, Allahabad, Srinagar, New Delhi, Surat, Ludhiana, Navi Mumbai, Ghaziabad, Bengaluru, Indore, Nagpur, Thane, Agra, Meerut, Ranchi. The delivery feasibility and charges may be varying, hence for them please check with the particular seller or store.	1

Table 10: A selection of substrings generated by XL-ORIGINAL with no prompting (and top- k with $k=50$) that were identified by EXACTSUBSTR as being in C4 multiple times. The number of times each substring was found in C4 is given. We observe that most memorized generations tend to be from advertisements.

RealNews Url	# Total	Frac Dups	C4 Url	# Total	Frac Dups
medicalnewstoday.com.	12	1.00	hairtechkearney.com	4883	1
dodbuzz.com	301	0.99	keywordsking.com	1786	1
undertheradar.military.com	187	0.97	sydneysitalianfruitshops.online	1178	1
q.usatoday.com	33	0.94	moewiki.usamimi.info	1001	1
ad-test.thirdage.com	354	0.94	swarovskijewelryoutlet.org	984	1
amp.nymag.com	15	0.93	forzadurto.org	980	1
citizenwire.com	1022	0.93	producerati.com	971	1
paycheck-chronicles.military.com	363	0.92	sourceryforge.org	908	1
product-reviews.net	73403	0.92	heavenz-kitchen.com	876	1
kitup.military.com	196	0.92	little-eclipse.com	822	1
gcaptain.com	33903	0.92	walops.com	819	1
dev.screenrant.com	70	0.91	16thstlaunderland.com	713	1
live.swissinfo.ch	66	0.91	theroyalstarinfo.com	696	1
news.theepochtimes.com	82	0.87	code4kt.com	684	1
opinion.toledoblade.com	986	0.87	nflfalconsjerseys.us	682	1
cdn.moneytalksnews.com	121	0.86	quiltingbeeshop.com	676	1
amp.fox23.com	14	0.86	ulifeinsurancemiami.com	675	1
sales.rollingstone.com	20	0.85	wowkeyword.com	673	1
ftp.screenrant.com	20	0.85	taspetro.com	671	1

Table 11: On the left, we show the URLs that had the greatest proportion of examples marked as near-duplicates by NEARDUP(filtered to URLs which occurred at least 10 times). On the right, we show the 20 most frequent URLs in C4 for which all examples were marked as near-duplicates by NEARDUP.

Training Dataset:	C4-ORIGINAL		C4-NEARDUP		C4-EXACTSUBSTR	
Epoch:	1	2	1	2	1	2
No prompt	1.93%	1.57%	0.19%	0.26%	0.14%	0.17%
Duplicate Train Prompts	35.88%	34.34%	3.34%	3.15%	5.71%	4.67%
Unique Train Prompt	0.42%	0.41%	0.42%	0.41%	0.22%	0.23%
Duplicate Test Prompt	16.27%	15.32%	1.61%	1.52%	0.34%	0.25%
Unique Test Prompt	0.25%	0.22%	0.21%	0.23%	0.03%	0.08%

Table 12: Percentage of tokens in 100k generations that were part of memorized substring according to EXACT-SUBSTR. Models trained with approximate or exact deduplication have $10\times$ less memorization than the model trained on the original (non-deduplicated) dataset.

	Final train set size in tokens			Final train set size in examples		
	ORIGINAL	NEARDUP	EXACTSUBSTR	ORIGINAL	NEARDUP	EXACTSUBSTR
C4	177.3B	173.7B	165.4B	364.87M	350.48M	350.48M
Real News	24.7B	22.4B	20.1B	31.16M	28.39M	28.39M
LM1B	1.0B	0.94B	0.90B	30.30M	29.87M	30.16M
Wiki40B	2.25B	2.24B	2.19B	2.93M	2.91M	2.93M

Table 13: Each row shows the size in tokens (according to our 50k BPE vocab) and in examples of a train set in its original form, with NEARDUP deduplication, and with EXACTSUBSTR deduplication.